

Spring 2025 Traders Project

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Our project explores a data-driven trading strategy based on Bollinger Bands and the concept of mean reversion, designed to identify buy and sell signals in volatile, sentiment-driven assets. The core idea stems from the observation that prices often oscillate within a range, reverting to their average over time—a phenomenon particularly exploitable in sideways or consolidating markets. To minimize false positives, we integrate confirmation signals such as volume spikes, RSI divergence, and candlestick reversals. This strategy specifically targets two assets: AMD, a semiconductor stock known for sharp sentiment swings, and XLE, an energy sector ETF influenced by oil volatility and cyclical market behavior. Using historical daily data from January 2020 to January 2024, we constructed a systematic backtest using open, high, low, close, and volume data.

The methodology centers on identifying oversold conditions where both AMD and XLE fall below their lower Bollinger Bands—serving as a buy trigger. Positions are sold when either asset closes above its middle Bollinger Band (20-day SMA). Each trade allocates 60% of the available capital, with a starting portfolio value of \$100,000. To enhance signal reliability and handle market noise, we incorporated Kalman Filters, which dynamically estimate underlying trends and support more precise entry and exit decisions. These filters, commonly used in control systems and finance, are particularly effective for smoothing volatile price data while maintaining responsiveness to regime shifts.

Backtesting results showed 29 trades over the five-year period, yielding an ending capital of \$283,940.28 and a total return of 183.94%. The strategy achieved a Sharpe Ratio of 1.129, signifying adequate risk-adjusted returns. These results validate our thesis that short-term overreactions in specific assets can be systematically exploited with proper signal filtering and risk management.

In conclusion, our project demonstrates the viability of a Bollinger Band-based mean reversion strategy when combined with market sentiment tools and filtering techniques. The model offers stable returns with limited downside exposure and serves as a foundational prototype for more advanced strategies. Future enhancements may include optimizing parameters, applying the strategy across multiple sectors, or integrating machine learning to dynamically adjust thresholds in real time.